

When Does Cross-City Graph Transfer Help? Few-Shot Bike-Sharing Demand Forecasting Across Heterogeneous European Systems

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Abstract

Cross-city transfer may reduce the target supervision needed for station-level bike-sharing forecasting, but differences in demand scale, station layout and observation coverage can cause negative transfer. Standard within-city models on fixed graphs do not establish whether shared parameters remain useful on unseen station networks as the target-label adaptation budget changes. We study next-hour departures across twelve European systems, reserving eight cities for development and source fitting and four for held-out evaluation. A shared Graph-GRU is compared with a matched Vanilla-GRU, deterministic baselines, pooled training, ordinary source pretraining and GRL-based multi-source domain-invariant pretraining. Target supervision ranges from parameter zero-shot to one, seven and thirty days and the full permitted pre-test history. Parameter zero-shot excludes target labels from fitting, adaptation and model selection; causal target histories remain forecasting inputs. Coverage masks separate observed zeros from unavailable station-hours. Development fixes the architecture and adaptation policy; evaluation uses five seeds and paired temporal-bootstrap intervals. Across four held-out cities, ordinary transfer improved over target-only Graph-GRU at every nonzero budget, with paired equal-city MAE gains of 0.119–0.195. Graph propagation helped at one day but not at larger budgets, while the evaluated GRL specification consistently underperformed ordinary transfer. Pooled training achieved standalone MAE 0.869 at seven days; target historical averaging was best among registered methods at thirty days and full history. These findings concern the four observed city-periods.

Introduction

Station-level bike-sharing demand forecasts support rebalancing, dock allocation and maintenance. Cross-city transfer could reduce the target supervision needed to predict next-hour departures. Its benefit must be established against local models trained on the same limited target supervision.

This is a *parameter zero-shot and limited-supervision* setting, rather than a history-free cold start. Parameter zero-shot means that no target labels are used for parameter fitting, adaptation or model selection. Strictly causal recent

target-city observations, including demand histories and a weekly lag, may still serve as forecasting inputs. The controlled resource is therefore the *target supervision budget*, not the existence of target observations at inference.

Transfer is challenging because cities differ in network size, station spacing and demand scale. The twelve systems studied here come from the European Bike-Sharing Dataset (Waldner et al. 2025). Patterns learned in a dense, busy system may be inappropriate elsewhere, producing negative transfer: performance worse than target-only training (Rosenstein et al. 2005; Pan and Yang 2010). Incomplete observation creates a second challenge. Change-triggered station feeds make the absence of a record insufficient evidence of zero demand. Confusing unavailable station-hours with verified zeros can distort both training and evaluation.

Graph-based forecasters model station interactions (Lin, He, and Peeta 2018; Chai, Wang, and Yang 2018). Cross-city transfer and meta-learning reuse source knowledge (Wang et al. 2019; Yao et al. 2019), while adversarial learning reduces domain information (Ganin and Lempitsky 2015). These approaches motivate testing when source knowledge helps in this heterogeneous station-level setting as target supervision increases.

We compare deterministic baselines, graph and graph-free recurrent models, pooled training, ordinary source pretraining and gradient-reversal-based multi-source pretraining. Budgets range from parameter zero-shot through one, seven and thirty days to the full permitted pre-test history. Eight development cities determine the fixed design; four held-out cities supply the final evaluation. Sequential transfer keeps source exposure constant while varying target adaptation.

The contribution is a controlled empirical benchmark, with no architectural novelty claim. Shared parameters support unseen station graphs; observation masks keep missingness explicit; matched comparisons separate graph effects, source pretraining and domain-invariance effects. Budget-dependent and city-level reporting is designed to reveal both helpful and negative transfer.

Related Work

Application Perspective

Station-level bike-sharing forecasting benefits from representing interactions across a station network. Lin, He, and Peeta (2018) learn station interactions jointly with demand prediction, while Chai, Wang, and Yang (2018) combine multiple graphs to represent spatial and functional relationships. These studies motivate spatial modeling. Our focus is the separate question of whether shared forecasting parameters remain useful across heterogeneous cities with limited target supervision.

Methodology Perspective

DCRNN combines diffusion on a graph with recurrent forecasting (Li et al. 2018); STGCN uses temporal convolutions (Yu, Yin, and Zhu 2018); and T-GCN couples graph convolution with a GRU (Zhao et al. 2020). Standard applications are typically instantiated on a fixed sensor graph, whereas our setting requires shared parameters that can be applied directly to unseen station graphs of different sizes. This is a requirement of our experimental setting, not a claim that these architectures fundamentally cannot transfer.

Cross-city prediction has used region matching to guide parameter reuse (Wang et al. 2019) and meta-learning to support rapid adaptation (Yao et al. 2019). Domain-adversarial learning instead trains representations against a domain discriminator (Ganin and Lempitsky 2015). We apply that principle across source cities, without using the final target as an unlabeled pretraining domain. Because transfer can also hurt (Rosenstein et al. 2005; Pan and Yang 2010), ordinary and adversarial source pretraining are compared under identical target adaptation. The benchmark examines how transfer gains change with the target-label adaptation budget, alongside a matched graph ablation. Neither the Graph-GRU backbone nor gradient reversal is presented as a new architecture.

Methodology

Problem Setting and Inputs

For a city with N stations, let $y_{s,t} \in \mathbb{Z}_{\geq 0}$ denote hourly departures and $m_{s,t} \in \{0, 1\}$ their observability. Given information available at forecast origin t , we predict

$$\hat{y}_{s,t+1} = f_{\theta}(\mathbf{X}_t^{\text{hist}}, \mathbf{X}_t^{\text{week}}, \mathbf{X}_t^{\text{static}}, \mathbf{x}_t^{\text{cal}})_s. \quad (1)$$

No learned parameter is indexed by station or city. The same weights can therefore be instantiated on an unseen graph with a different station count. *Parameter zero-shot* excludes target labels from fitting, adaptation and model selection, while permitting strictly causal target observations as inputs.

Neural models share the same features. The history tensor $\mathbf{X}^{\text{hist}} \in \mathbb{R}^{B \times 24 \times N \times 2}$ contains 24 hourly demand lags in $\log(1 + y)$ space, each paired with its observation indicator. The weekly tensor $\mathbf{X}^{\text{week}} \in \mathbb{R}^{B \times N \times 2}$ carries the same pair at $t - 168$ h. Static features comprise $\log(1 + \text{capacity})$ and a validity flag. Six calendar features encode local hour and weekday with sine–cosine pairs, weekend status and UTC offset divided by 12. Weather, holidays, events, identifiers and dock occupancy are excluded. The absence of

explicit identifiers does not prevent city information being inferred from these features. An unavailable demand input is encoded as $(0, 0)$ (numeric value, indicator), whereas an observed zero is $(0, 1)$; positive observations use $(\log(1 + y), 1)$. All lags are elapsed UTC hours.

Departure Aggregation and Coverage

Hourly departures are counts of valid, city-matched inferred trips grouped by departure station and start time in the half-open UTC interval $[u, v)$, where $v = u + 1$ h. Duplicate and unmatched records are excluded. These inferred movements are not operator transaction logs. Counts are joined to a fixed station-hour grid; an absent trip row alone never establishes zero demand.

Coverage is certified using status evidence available by cutoff a . For each station, retain only events timestamped at or before a . The hour start must lie within that prefix’s first and last events. Let p be the latest event at or before u and n the earliest at or after v , with $n \leq a$. Both must exist, with $u - p \leq 12$ h and $n - v \leq 12$ h. The city gate additionally requires at least half of the fixed-roster stations whose prefix lifetimes cover u to satisfy this bracket; an empty denominator fails. No same-hour event is required. These conditions define observability.

Predictor masks use evidence available at the forecast origin and require $v \leq a$; fitting targets are certified at the fitting cutoff. Later retrospective evaluation evidence cannot alter predictors or fitted models. When the target mask is one, no matched departure is a coverage-qualified zero. Otherwise the target is unavailable and excluded from loss and metrics, even when a positive raw count exists. This operational definition of a verified zero establishes coverage, not the absence of all latent demand.

Geographic Graph and Graph-GRU

For each city, haversine distances d_{ij} define a geographic k -nearest neighbor graph. Directed edges are symmetrized by union. With retained edge set $\mathcal{E}$ and bandwidth $\rho = \text{median}_{(i,j) \in \mathcal{E}} d_{ij}$, let $W_{ij} = \exp[-(d_{ij}/\rho)^2]$ on retained edges, zero otherwise, and $W_{ii} = 1$. Symmetric normalization (Kipf and Welling 2017) gives

$$\hat{A} = \tilde{D}^{-1/2} W \tilde{D}^{-1/2}, \quad \tilde{D}_{ii} = \sum_j W_{ij}. \quad (2)$$

Development selection fixes $k = 4$. The median bandwidth adapts to geographic scale without per-city tuning.

Graph propagation precedes each recurrent gate (Cho et al. 2014). With hidden state $\mathbf{H}_t \in \mathbb{R}^{N \times H}$, input $\mathbf{X}_t \in \mathbb{R}^{N \times 2}$ and $\mathbf{Z}_t = [\mathbf{X}_t \| \mathbf{H}_{t-1}]$,

$$\mathbf{r}_t = \sigma(\hat{A} \mathbf{Z}_t \mathbf{W}_r + \mathbf{b}_r), \quad (3)$$

$$\mathbf{u}_t = \sigma(\hat{A} \mathbf{Z}_t \mathbf{W}_u + \mathbf{b}_u), \quad (4)$$

$$\mathbf{c}_t = \tanh(\hat{A} [\mathbf{X}_t \| \mathbf{r}_t \odot \mathbf{H}_{t-1}] \mathbf{W}_c + \mathbf{b}_c), \quad (5)$$

$$\mathbf{H}_t = \mathbf{u}_t \odot \mathbf{H}_{t-1} + (1 - \mathbf{u}_t) \odot \mathbf{c}_t. \quad (6)$$

Here σ is the sigmoid and $\odot$ is elementwise multiplication. Reset and update gates share the propagated tensor $\hat{A} \mathbf{Z}_t$. The recurrence processes the 24 lags chronologically. The final

state and ten-dimensional weekly, static and calendar context enter a non-negative prediction head. Hidden width is $H = 32$, with no recurrent dropout. The matched Vanilla-GRU substitutes the identity for $\hat{A}$ and retains the same width, context, head and training protocol. This is the graph ablation.

Forecasting Objective

Training uses masked absolute error in $\log(1 + y)$ space:

$$\mathcal{L}_{\text{fc}} = \frac{\sum_{s,t} m_{s,t} |\hat{p}_{s,t} - \log(1 + y_{s,t})|}{\sum_{s,t} m_{s,t}}. \quad (7)$$

The non-negative output $\hat{p}$ is converted to counts by $\hat{y} = \max(0, \exp(\hat{p}) - 1)$. Joint training averages loss within each city and then equally across cities. Unavailable target values are excluded, rather than treated as zero-demand observations. Output space and loss were selected on development cities.

Transfer Regimes and Gradient Reversal

Four deterministic baselines are persistence, seasonal naive, a source historical average and a target hour-of-week historical average. The last requires the corresponding target history for estimation and is unavailable at zero target supervision. Target-only Vanilla-GRU and Graph-GRU start from fresh initialization. Pooled Graph-GRU jointly uses the fixed source exposure and target-budget examples. Sequential transfer instead pretrains on sources and then updates all encoder and prediction-head parameters on the target budget, without source replay. Target-only and pooled neural models are defined only for non-zero budgets.

The adversarial branch adds a city discriminator during source pretraining (Ganin and Lempitsky 2015). Let $\mathbf{h}_i$ be the mask-weighted mean of the final station hidden states for source example i , and c_i its city class. For discriminator probabilities D_ϕ , city cross-entropy is

$$\mathcal{L}_{\text{city}} = -\frac{1}{M} \sum_{i=1}^M \log D_\phi(\mathbf{h}_i)_{c_i}. \quad (8)$$

The gradient reversal layer satisfies

$$R_\lambda(\mathbf{h}) = \mathbf{h}, \quad \left. \frac{\partial R_\lambda}{\partial \mathbf{h}} \right|_{\text{backward}} = -\lambda I. \quad (9)$$

For forward loss $\mathcal{L}_{\text{fc}} + \mathcal{L}_{\text{city}}$, encoder parameters θ receive

$$g_\theta = \nabla_\theta \mathcal{L}_{\text{fc}} - \lambda \nabla_\theta \mathcal{L}_{\text{city}}. \quad (10)$$

The discriminator minimizes city cross-entropy; the forecasting head minimizes forecasting loss. The $32 \rightarrow 64 \rightarrow C$ discriminator uses ReLU and dropout 0.1. There are $C = 7$ source classes in each development fold and $C = 8$ in the final source refit. Development selected constant $\lambda = 0.5$. The discriminator is discarded before target adaptation; final target cities never become discriminator classes. This is multi-source domain-invariant pretraining, not target-domain adversarial adaptation. Comparing it with ordinary source pretraining under identical adaptation is the domain-invariance ablation.

System(s)	Role	Stations
All 12 systems	8 development, 4 target	30–146
Mannheim	Target	78
Innsbruck	Target	45
Glasgow	Target	93
Split	Target	60

Table 1: Dataset statistics: overall station-count range and fixed target evaluation rosters. Across the twelve systems, mean departures per station-day range from 7.9 to 101.9.

Experiments and Results

Dataset and Split

The high-confidence subset contains twelve European Bike-Sharing Dataset systems (Waldner et al. 2025). Development and source fitting use Bilbao, Cardiff, Freiburg, Göteborg, Dortmund, Heidelberg, Marburg and Gießen. Four leave-one-city-out selection folds use Bilbao, Cardiff, Freiburg and Göteborg as pseudo-targets, with the other seven development cities as sources. Mannheim, Innsbruck, Glasgow and Split are held-out targets. Table 1 summarizes the systems and target rosters.

Development source fitting begins on 29 August 2022 at 05:00 UTC; selection ends before 16 April 2023 at 22:00 UTC. Final source refits and target adaptation use permitted pre-cutoff history. A 593-hour embargo separates the fitting cutoff from the final interval, 11 May 2023 at 15:00 UTC to 15 July 2023 at 20:00 UTC, comprising 1,565 forecast origins. All architecture, graph, loss and optimization choices are fixed using development cities; final target results cannot reopen selection.

Preprocessing

Hourly inferred departures and their coverage indicators follow the aggregation described in Methodology. Unavailable station-hours are excluded from training targets and all metrics. The benchmark assumes each reconstructed count becomes available once its hour ends, so subsequent forecast origins use only causal counts. This idealized count feed is not a measured production-ingestion replay. Static metadata use a fixed retrospective station roster because publication timestamps are unavailable. At 0d, target observations remain inputs but are not used for fitting, adaptation or model selection.

Implementation Details

The PyTorch implementation uses AdamW with $\beta = (0.9, 0.999)$, $\epsilon = 10^{-8}$, weight decay 10^{-4} and global gradient clipping at norm 1. Learning rates are 10^{-3} for source updates and 2×10^{-4} for target updates. Each update samples 16 eligible forecast hours with replacement from one city and includes its full station set. Source schedules balance cities exactly. Development fixes log-space masked MAE, $k = 4$, $H = 32$, zero recurrent dropout and $\lambda = 0.5$; GRL selection uses post-seven-day-adaptation count-space MAE on the four pseudo-target folds.

Five fixed seeds are 17, 29, 43, 71, 101. Source pretraining uses 12,000 updates; target-only fitting and adaptation use Table 3’s counts. Sequential adaptation resets AdamW state and has no source replay. Pooled training interleaves exactly the same source and target update counts, switching learning rate by domain while retaining optimizer state. Training uses CUDA with deterministic algorithms and retains the last fixed-step checkpoint, without early stopping or target-based model selection.

Metrics

Let Ω be the common set of valid matched station-hours and $e_{s,t} = y_{s,t} - \hat{y}_{s,t}$. Count-space metrics are

$$\text{MAE} = \frac{1}{|\Omega|} \sum_{(s,t) \in \Omega} |e_{s,t}|, \quad (11)$$

$$\text{RMSE} = \sqrt{\frac{1}{|\Omega|} \sum_{(s,t) \in \Omega} e_{s,t}^2}, \quad (12)$$

$$\text{WAPE} = \frac{\sum_{(s,t) \in \Omega} |e_{s,t}|}{\sum_{(s,t) \in \Omega} |y_{s,t}|}. \quad (13)$$

For $\Omega_s = \{t : (s, t) \in \Omega\}$ and $S_{24} = \{s : |\Omega_s| \geq 24\}$, station-macro MAE is

$$\text{MAE}_{\text{station}} = \frac{1}{|S_{24}|} \sum_{s \in S_{24}} \frac{1}{|\Omega_s|} \sum_{t \in \Omega_s} |e_{s,t}|. \quad (14)$$

WAPE is a ratio, not a count error; zero denominators make it undefined. MAE is primary. Metrics are computed per city and seed, then summarized by the seed mean and population standard deviation; headline results weight the four cities equally.

Paired temporal bootstrap intervals use nine complete 168-hour blocks and one terminal 53-hour block per city. Each of 2,000 replicates samples ten block IDs with replacement, keeping any selected terminal block intact. Draws are shared across methods, budgets, seeds and metrics. The 2.5th and 97.5th percentiles use linear interpolation; undefined replicates are not redrawn. These intervals describe temporal uncertainty, not variation across a population of European cities.

Held-Out Results and Ablations

Table 2 reports standalone MAE. Ordinary source pretraining, seasonal naive and pooled Graph-GRU are best at 0d, 1d and 7d, respectively. Target historical averaging has the lowest registered standalone MAE at 30d (0.835273) and Full (0.807435); ordinary transfer is therefore not uniformly best across all methods.

Controlled comparisons instead use fixed matched prediction intersections. For the evaluated method and its specified reference, define

$$G_b = \text{MAE}_b^{\text{reference}} - \text{MAE}_b^{\text{method}}. \quad (15)$$

Positive values favor the evaluated method. Each term is evaluated on the same paired cohort, so the gain need not

Method	0d	1d	7d	30d	Full
Source historical average	1.072289	—	—	—	—
Target historical average	—	0.976402	0.941287	0.835273	0.807435
Persistence	—	—	—	—	—
Seasonal naive	0.878195	0.878195	0.878195	0.878195	0.878195
Target-only Vanilla-GRU	—	1.111835	0.996551	0.970196	0.956180
Target-only Graph-GRU	—	1.084970	0.996926	0.992346	0.991063
Pooled Graph-GRU	—	0.888162	0.869285	0.879580	0.906724
Ordinary source-pretrained Graph-GRU	0.874405	0.889892	0.873627	0.872275	0.871797
GRL-based source-pretrained Graph-GRU	0.940250	0.961752	0.943779	0.936276	0.936954

Table 2: Standalone equal-city macro count-space MAE across four held-out cities (lower is better). Bold marks the lowest estimable value across all rows in each column. “—” denotes an unavailable or inapplicable result; persistence is unavailable throughout. Learned-model entries aggregate five seeds. These standalone cohorts differ from the fixed paired intersections used for controlled gains; subtracting table cells does not recover those gains.

equal a difference of standalone table cells. Brackets below denote frozen 95% temporal-bootstrap intervals.

Graph ablation. Target Graph-GRU versus matched Vanilla-GRU gives:

Budget	Gain	95% interval
1d	+0.026866	[+0.024876, +0.029131]
7d	−0.000375	[−0.000571, −0.000170]
30d	−0.022149	[−0.024539, −0.019483]
Full	−0.034883	[−0.038920, −0.030387]

Spatial propagation helps at the lowest supervision budget, is nearly neutral but slightly worse at 7d, and underperforms the graph-free control at larger budgets in this setup.

Ordinary transfer. Source adaptation versus target-only Graph-GRU gives:

Budget	Gain	95% interval
1d	+0.195077	[+0.186929, +0.201726]
7d	+0.123300	[+0.112944, +0.133392]
30d	+0.120071	[+0.109447, +0.130570]
Full	+0.119266	[+0.108075, +0.130089]

Adaptation improves on target-only Graph-GRU at every nonzero budget. At 0d, seasonal naive is the registered deterministic reference (standalone MAE 0.878195); ordinary source transfer has controlled gain −0.007562 [−0.011856, −0.003193]. It does not beat that reference on the matched cohort, despite its lower standalone MAE of 0.874405.

Domain-invariance ablation. GRL-based multi-source domain-invariant pretraining versus ordinary pretraining gives −0.065845 [−0.070791, −0.060709] at 0d. Under identical adaptation:

Target budget	0d	1d	7d	30d	Full
Target updates	0	100	300	600	1,200

Table 3: Sequential transfer schedule. Every budget uses the same 12,000-update source pretraining exposure. Full denotes the permitted pre-test target history; 0d denotes parameter zero-shot.

GRL versus ordinary adaptation

Budget	Gain	95% interval
1d	−0.071859	[−0.077839, −0.065548]
7d	−0.070153	[−0.075086, −0.064959]
30d	−0.064002	[−0.068865, −0.058822]
Full	−0.065156	[−0.070015, −0.059976]

The evaluated GRL specification did not improve ordinary transfer. This does not establish that domain-invariant learning is generally ineffective.

Pooled versus sequential. With ordinary sequential adaptation as reference, pooled Graph-GRU gives:

Budget	Gain	95% interval
1d	+0.001730	[−0.003138, +0.005995]
7d	+0.004342	[+0.002177, +0.006312]
30d	−0.007305	[−0.010147, −0.004369]
Full	−0.034927	[−0.041053, −0.028319]

Pooling is competitive under scarce supervision, with an interval spanning zero at 1d and a positive gain at 7d; sequential adaptation is better at larger budgets.

City heterogeneity. Standalone zero-shot ordinary transfer improves on seasonal naive in Glasgow and Split but not Mannheim and Innsbruck. For example, Mannheim has MAE 0.9774 versus seasonal naive’s 0.9639, whereas Split has 0.7861 versus 0.8074. At Full in Innsbruck, pooled MAE is 1.2321 versus 1.1020 for ordinary adaptation, illustrating that more target supervision does not uniformly favor pooling. These are descriptive city-level standalone comparisons, not paired effects. All findings concern four observed city-periods; the temporal intervals do not establish population-level statistical significance across European cities.

Code and data. Code and configurations:

<https://github.com/melisuresin-oss/>

bike-sharing-transfer-learning.

Dataset (Waldner et al. 2025):

<https://github.com/tumftm/european-bike-sharing-dataset>.

Presentation:

https://drive.google.com/drive/folders/13au-S79TYy3v8xBDL3Z2-2A652nBlmRS?usp=drive_link.

drive_link.

Conclusion

Cross-city source pretraining improved over target-only Graph-GRU at every nonzero target supervision budget, with paired MAE gains of 0.119–0.195. Graph propagation itself helped only at the one-day budget and did not improve the matched Vanilla-GRU at larger budgets. The evaluated GRL-based multi-source domain-invariant pretraining consistently underperformed ordinary source pretraining. Ordinary parameter-zero-shot transfer also failed to beat seasonal naive on the controlled matched cohort. Effects varied by city and budget: pooling helped at seven days but lost to sequential adaptation with larger histories, while target historical averaging achieved the lowest standalone MAE at thirty days and Full. These conclusions are conditional on inferred departures, an idealized count feed, retrospective station metadata and four held-out city-periods; they do not establish universal transfer superiority or population-level significance.

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